

Assignment 2.1: Overview of Peripherals Theory

Task 1: Create a table listing 3 types of printers, 2 types of scanners, and 2 types of webcams, and for each, write one real-life use case (for example: 'Inkjet printer – printing college assignments').

Table 1. Printers, Scanners and Webcams

Device Category	Type	Real-Life Use Case
Printer	Inkjet Printer	Printing photographs. No Larger quantity.
Printer	Laser Printer	Printing larger quantity. Like Office documents.
Printer	Dot Matrix Printer	Multi-part carbon copies, low-cost industrial printing
Scanner	Flatbed Scanner	Scanning certificates, photographs, and important documents
Scanner	Handheld Scanner	Scanning barcodes. Mainly use in stores to scan product's barcode.
Webcam	USB Webcam	Attending online classes and video meeting. Like Google meet webcam.
Webcam	Wi-Fi Webcam	Monitoring a home or office. Which can access remotely via Internet

Task 2: Draw a labelled diagram showing how a typical home printer connects to a laptop or PC, including the connection interface used (USB, Wi-Fi, or Bluetooth).

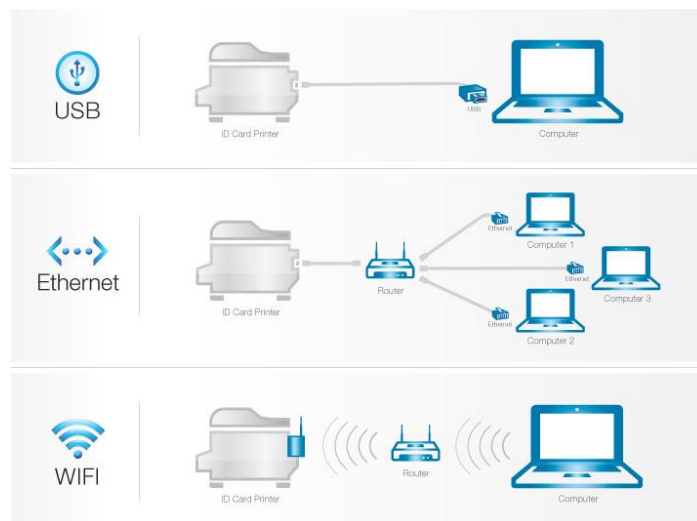


Fig 1. Printer connection via different media

Task 3: List three different connection interfaces (such as USB, Wi-Fi, Bluetooth) used by peripherals like printers, scanners, and webcams, and for each, name one advantage and one disadvantage. Hint: Think about speed, convenience, and compatibility.

USB:

Advantages	Disadvantages
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Fast data transfer speed	Requires a physical cable
Stable and reliable connection	Limited by cable length
Easy to set up	Uses a USB port on the computer

Wi-Fi:

Advantages	Disadvantages
No cables required	Requires a Wi-Fi network
Multiple devices can connect to the same printer or scanner.	Connection may become slow or unstable if the signal is weak
Allows use from anywhere within Wi-Fi range	Setup can be more complex than USB

Bluetooth:

Advantages	Disadvantages
Wireless and easy to pair	Slower data transfer than USB
Low power consumption	Limited range
Suitable for short-distance connections	Not supported by all devices

Task 4: Pick any one device you use at home or college (printer, scanner, or webcam). Write step-by-step instructions for how you would connect it to your computer, including what cable or wireless method you would use and any setup steps required.

USB Printer:

1. Place the printer near your computer and connect it to a power source.
2. Turn on the printer using the power button.
3. Connect the **USB cable** from the printer to a USB port on the computer.
4. Wait for the computer to detect the printer automatically.
5. If required, install the printer driver from the manufacturer's CD or website.
6. Open **Settings → Bluetooth & devices → Printers & scanners** (Windows) and check if the printer appears in the list.
7. Print a **test page** to make sure the printer is working properly.

Wi – Fi Printer:

1. Turn on the printer and enable **Wi-Fi** from the printer's settings.
2. Connect the printer to your home or college Wi-Fi network.
3. On your computer, open **Settings → Bluetooth & devices → Printers & scanners**.
4. Click **Add device** and select your printer from the list.
5. Install the required drivers if prompted.
6. Print a test page to confirm the connection.